

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

SCSR2213 NETWORK COMMUNICATIONS

Task#2

Group 4: Linux
SECTION: 5
Lecturer: Dr. Adnan Khalid

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Primenlary Analysis:

Introduction

To understand the requirements of our networking lab project, we decided to prepare and answer a list of 11 questions. These questions were divided between general understanding and specific needs (hardware, software, bandwidth, and security). Our findings helped us determine the technical and operational feasibility of the proposed design and were used to support the overall planning.

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Questions and Answers:

1. **What devices are needed for the Cisco networking lab?**

A comprehensive Cisco networking lab requires several key devices. Switches are essential for connecting multiple devices within a local area network (LAN), facilitating efficient communication and resource sharing. Routers are necessary for connecting multiple networks and directing data traffic between them. Additionally, wireless access points (WAPs) enable wireless connectivity for mobile devices. The inclusion of firewalls, such as the Cisco ASA series, is critical to secure network traffic and support training in cybersecurity. For network simulation and design, software like Cisco Packet Tracer provides an invaluable platform for students to practice and refine their networking skills.

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2. **What bandwidth is required to ensure smooth operation for the video conferencing room?**

To ensure seamless video conferencing, adequate bandwidth must be provisioned. The recommended bandwidth per participant in group video calls is 8 Mbps, with an additional 20% overhead considered for protocol headers and encapsulation. Given that the conferencing room accommodates 18 users, the minimum base bandwidth required would be 144 Mbps. Including the overhead, the total recommended bandwidth increases to approximately 173 Mbps, ensuring uninterrupted, high-quality conferencing experiences.

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3. **What are the pros and cons of wireless connectivity for the student lounge?**

Wireless connectivity in student lounges offers several advantages. It provides flexibility, allowing students to access the network from anywhere within the lounge. It also facilitates collaborative learning, enabling real-time file sharing and group work. Wireless systems are generally easier and faster to install than wired alternatives and can accommodate multiple devices simultaneously. However, wireless networks may experience performance degradation during periods of high usage, and if not properly secured, they are vulnerable to unauthorized access and cyber threats.

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4. **Why is it important for labs to be equipped with technology that aligns with the 4th Industrial Revolution (4IR) standards?**

Integrating 4IR technologies into educational labs is crucial to align academic training with evolving industry demands. Exposure to advanced technologies such as artificial intelligence (AI), robotics, and the Internet of Things (IoT) enhances students' technical competencies and prepares them for the modern workforce. Furthermore, working with 4IR tools fosters the development of critical thinking, problem-solving, and effective communication skills, all of which are essential for success in today's dynamic professional environments.

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5. **What types of user authentication methods are generally recommended for educational networks?**

To secure sensitive academic and administrative data, educational institutions are advised to implement robust authentication mechanisms. Single Sign-On (SSO) simplifies access management by allowing users to authenticate once and gain access to multiple systems. Multi-Factor Authentication (MFA) adds an additional security layer by requiring secondary verification, such as a one-time code sent to the user's mobile device, thereby significantly enhancing protection against unauthorized access.

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6. **What physical and environmental requirements are recommended for server rooms in educational institutions?**

Server rooms should adhere to strict physical and environmental standards to ensure optimal equipment performance and longevity. The ambient temperature should be maintained between 18°C and 27°C, ideally around 22°C. The relative humidity should be controlled between 45% and 50% to prevent both moisture-related corrosion and electrostatic discharge. Implementing hot and cold aisle containment and utilizing HVAC systems equipped with real-time monitoring sensors are also recommended to maintain stable and efficient operating conditions.

7. **Are there common practices for implementing Quality of Service in educational networks to prioritize traffic for learning and administrative applications?**

Educational networks commonly deploy Quality of Service (QoS) techniques to prioritize time-sensitive applications such as video conferencing, VoIP, and e-learning platforms. Technologies such as Differentiated Services (DiffServ) and Resource Reservation Protocol (RSVP) are utilized to classify and manage network traffic. Additionally, network devices are configured with multiple queues to differentiate service levels. Centralized QoS policy management tools, like Cisco's QoS Policy Manager, enable IT teams to monitor and enforce QoS configurations campus-wide.

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8. **How is Wi-Fi coverage typically managed across faculty buildings to ensure reliable connectivity for students and staff in all areas**?

Achieving reliable Wi-Fi coverage across faculty buildings involves conducting comprehensive site surveys to identify high-density usage areas such as classrooms, lecture halls, and laboratories. High-capacity access points are strategically deployed to maximize coverage and performance. A centrally managed Wi-Fi system allows for streamlined control over network security and bandwidth allocation. Utilizing a Network-as-a-Service (NaaS) model can further enhance efficiency by outsourcing maintenance and scaling services according to institutional needs.

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9. **How can we improve the security of the desired system?**

Enhancing system security involves proactively addressing insider threats, ensuring all devices and communications are encrypted, and conducting regular security audits and penetration tests to identify and remediate vulnerabilities before they can be exploited.

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10. **What are the backup plans we should consider to prevent data loss?**

To effectively mitigate the risk of data loss, institutions should implement routine data backups, secure hardware physically, educate staff on cybersecurity best practices, maintain clean and organized work environments, deploy antivirus and anti-malware solutions, encrypt sensitive information, apply software patches promptly, and enforce strict security and password policies. Additionally, Data Loss Prevention (DLP) tools should be utilized to monitor and protect sensitive data across networks and endpoints.

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11. **What are the steps needed to acquire the utmost efficiency of the system?**

Maximizing system efficiency requires a multi-faceted approach, including upgrading network infrastructure, optimizing system configurations, efficient traffic management, and implementing proactive monitoring and maintenance strategies to detect and resolve issues before they impact system performance.

*** (note: the answers provided above is supported by references provided in the appendix below)**

Feasibility Analysis

This report summarizes the feasibility of implementing our Cisco networking lab project across three aspects:

A. Technical Feasibility

- The hardware and software required are readily available and well-supported by our institution's IT team.
- The technical complexity is manageable and aligned with the skills of the target users and support team.

B. Operational Feasibility

- The lab supports 4IR technology and serves undergraduate and postgraduate students.
- It enhances learning through exposure to realistic and advanced networking environments.

C. Economic Feasibility

- While initial costs may be high, long-term benefits (skill development, hybrid learning support) justify the investment.
- Bandwidth and hardware requirements are reasonable given the lab's expected output.

Conclusion: The Cisco lab, conferencing room, and student lounge Wi-Fi are all feasible from technical, operational, and economic standpoints. This investment aligns with the university's vision to prepare students for mode

- **This is a list of resources used for this project:**

1) NetworkLessons.com. (n.d.). Recommended lab equipment for Cisco CCNA. https://networklessons.com/cisco/ccna-routing-switching-icnd1-100-105/recommended-lab-equipment-for-cisco-c cna "reference for question 1"

2) NetDevGroup. (n.d.). NETLAB+ Requirements for Cisco Networking Academies. https://www.netdevgroup.com/content/cnap/requirements/ "reference for question 2"

3) Thomas Conner. (2024, March 3). Video conferencing bandwidth requirements. Targeting Mantra. https://targetingmantra.com/video-conferencing-bandwidth-requirements/ "reference for question 3"

4) Howard. (2024, July 30). Transforming education: Unlocking the benefits of Wi-Fi for schools. FS Community. https://community.fs.com/article/transforming-education-unlocking-the-benefits-of-wifi-for-schools.html "reference for question 4"

5) Huzaifa Khan. (2023, June 7). Preparing students for the Fourth Industrial Revolution: The role of 4IR skills in education. LinkedIn. https://www.linkedin.com/pulse/preparing-students-fourth-industrial-revolution-role-4ir-huzaifa-khan "reference for question 5"

6) Williams, J. J. (2021, November 10). *The Four Steps to Better User Authentication for Educational Institutions.* *James John Williams*. https://digt.com/informationsecurity/tpost/drxyme2je1-the-four-steps-to-better-user-authentica "reference for question 6"

7) *Temperature & Humidity: Standards for Data Centre & Server Rooms - UK Data Centre Cleaning*. (2024, October 11). UK Data Centre Cleaning. https://www.ukdatacentercleaning.co.uk/temperature-and-humidity/ "reference for question 7"

8) Fritz, J. (2020, February 14). QoS on Campus. *Technology Solutions That Drive Education*. https://edtechmagazine.com/higher/article/2008/02/qos-campus "reference for question 8"

9) Spectra. (2024, October 1). *Campus-Wide Managed Wi-Fi: From Lecture Halls to Hostels - Spectra*. https://www.spectra.co/resources/blog/from-lecture-halls-to-hostels-ensuring-seamless-managed-wi-fi-o n-campus "reference for question 9"

10) *Nile*. (2024, August 6). *Wi-Fi for University and College Campuses: How to Set It Up - Nile.* https://nilesecure.com/network-as-a-service/wi-fi-for-university-and-college-campuses-how-to-set-it-up "reference for questions 10, 11"

Meeting minutes

DATE	20 April 2025 1600H
LOCATION	Online ((google meet)
AGENDA	Discussion for task 2
MEETING MC	GARRAH THABIT

ATTENDANCE

Name	Time	Reason for absent
LEE TUNG ERN	9:00 PM	-
GARRAH THABIT	9:00 PM	-
MOHAMMED ABDELRAHIM	9:00 PM	-
MOHAMMED ALSAKKAF	9:00 PM	-

MINUTES

Name	Task
GARRAH THABIT	✓ Conducting research to find answers for the assigned questions. ✓ Compiling a list of references and sources. ✓ Designing questions to ask the user.
LEE TUNG ERN	✓ Writing a preliminary feasibility report.
MOHAMMED ALSAKKAF	✓ Creating a table aligned with the evaluation rubrics.
MOHAMMED ABDELRAHIM	✓ Drafting a report to document the meeting discussions. ✓ Developing a table that outlines the task requirements.

Projected Marks

TASK 1		
ITEM	**MARKS**	**EXPECTED MARKS**
Questions are appropriate to project and beneficial for better understanding		2
Questions are answered correctly and appropriately		2
Questions are researched through FC representative and reputable sources.		1
Answers are correctly referenced (with reference and citation)		2
Feasibility answer		1
Feasibility reasoning is logical and appropriate.		2
TOTAL		10